# Particle Physics 2: The Standard Model
# Lecture 8: The Higgs Mechanism

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# Chapter 1

# The Higgs Mechanism: How a Gauge Boson Acquires Mass

We now join the two ideas developed in the previous lecture. A complex scalar field with a broken global $U(1)$ symmetry contains a massive radial oscillation and a massless angular oscillation. A locally invariant $U(1)$ theory contains a massless gauge field. When the scalar is charged and its equilibrium magnitude is nonzero, these statements cannot simply be added together. The angular mode and the gauge field combine: the gauge field becomes massive, the apparent Goldstone boson ceases to be an independent particle, and the radial Higgs mode remains.

The photon will be our model gauge boson, but not our physical conclusion. Electromagnetic $U(1)$ is not broken in the vacuum, so the real photon is massless. The same mechanism acts on the $W^{\pm}$ and $Z$ in the electroweak theory, where the gauge group and scalar multiplet are more complicated. The abelian model exposes the mechanism without hiding it under group theory.

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**Question from the audience.** Did the previous discussion mean that the Higgs mechanism gives a mass to the real photon?[a]

**Response.** No. We ask what would happen *if* the $U(1)$ associated with the photon were in a broken phase. That hypothetical photon would become massive. The real electromagnetic vacuum is not in that phase; the application to particle physics is instead the mass of the $W$ and $Z$.

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[a] Lecture timestamp: 00:00:23.

The symmetry associated with electric-charge conservation is the phase rotation

$$\phi(x) \longmapsto e^{i\theta}\phi(x), \tag{1.1}$$

an internal rotation called $U(1)$. A superconductor provides a real analogue of the broken phase. Inside it, the electromagnetic field has a finite penetration depth and behaves in important respects like a massive field. Outside it, the ordinary massless photon reappears.

## 1.1 What Mass Means for a Field

### 1.1.1 Dispersion and the zero-momentum gap

It is worth deciding precisely what we mean by mass before introducing the Higgs field. For a wave,

$$E = \hbar\omega, \qquad p = \hbar k, \tag{1.2}$$

so an $(E, p)$ graph and an $(\omega, k)$ graph contain the same information up to factors of $\hbar$. In empty space a massless relativistic excitation obeys

$$E = c|p|, \qquad \omega = c|k|. \tag{1.3}$$

The two signs of $k$ distinguish opposite directions of propagation. The essential point is that $\omega$ tends to zero with $k$.

A massive relativistic particle obeys

$$E^2 = c^2p^2 + m^2c^4. \tag{1.4}$$

In units $c = \hbar = 1$,

$$\omega^2 = k^2 + m^2. \tag{1.5}$$

At $k = 0$, the energy is $E = mc^2$, or simply $m$ in natural units. Thus mass is the nonzero energy gap of the zero-momentum mode.

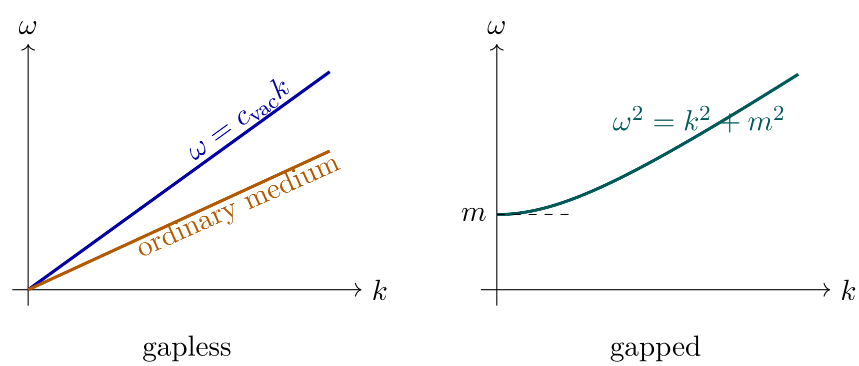


Figure 1.1: Changing the propagation speed changes the slope of a gapless dispersion relation. Giving the excitation a mass opens a nonzero gap at $k = 0$.

An ordinary transparent material may lower a wave's phase velocity without creating this relativistic mass gap. At a boundary the incident frequency is fixed, while the wave number adjusts to the medium's dispersion relation. Changing the slope is not the same as lifting the intercept. [1]

[1]Editorial clarification: A material can have several electromagnetic branches, and a plasma or resonant medium can possess a nonzero plasma-frequency gap. The lecture's comparison is specifically between an ordinary refractive branch and the superconducting response. More generally, gaplessness means $\omega(k) \to 0$ as $k \to 0$; exact linearity is not required in every nonrelativistic medium.

**Question from the audience.** Does a photon acquire mass whenever it travels through matter, for example through a prism?[a]

**Response.** No. Refraction can change the relation between frequency and wave number, and therefore the propagation speed, while the branch still passes through $\omega = 0$ at $k = 0$. A superconducting phase instead gives the electromagnetic response a finite inverse penetration length, the analogue of a mass.

[a]Lecture timestamp: 00:03:02.

**Question from the audience.** For light of a fixed frequency entering glass, is it the wave number that changes?[a]

**Response.** Yes. Time-translation invariance at a stationary interface keeps the frequency fixed. Because the dispersion curve has a different slope in the glass, the same $\omega$ corresponds to a different $k$, and therefore a different wavelength.

[a]Lecture timestamp: 00:09:20.

The slope

$$v_g = \frac{d\omega}{dk} \tag{1.6}$$

is the group velocity of a wave packet. If that slope varies with $k$, different Fourier components move at different speeds and the packet disperses. This is why the relation $\omega(k)$ is called the *dispersion relation*.

**Question from the audience.** What is the name of the curve, and what does its slope represent?[a]

**Response.** The curve is the dispersion relation. Its derivative $d\omega/dk$ is the group velocity. A constant slope gives the same velocity to all wavelengths; a varying slope makes a wave packet spread. For the present purpose, the decisive distinction is whether the curve reaches the origin or has a gap.

[a]Lecture timestamp: 00:11:02.

### 1.1.2 A homogeneous oscillation is a particle at rest

The mode $k = 0$ has infinite wavelength: the field is displaced by the same amount everywhere. Release a homogeneous displacement. If the field oscillates with a nonzero frequency, the corresponding quantum has a nonzero rest energy and is massive. If there is no restoring force in that direction, the frequency vanishes and the mode is massless.

This statement is classical. It does not require spatially separated particles to be in a quantum superposition. A homogeneous classical field configuration is simply one field value repeated throughout space.

**Question from the audience.** If an entire field is shifted together, is that the same as quantum entanglement?[a]

**Response.** No. A homogeneous classical field is one classical configuration. Entanglement correlates distinct quantum alternatives, such as up–down and down–up spin configurations. Preparing the same field value at many points is not by itself entanglement, nor does it imply an instantaneous causal response.

[a]Lecture timestamp: 00:10:04.

For a canonically normalized real field,

$$\mathcal{L} = \frac{1}{2}\partial_\mu\varphi\,\partial^\mu\varphi - V(\varphi), \tag{1.7}$$

expand about an equilibrium $\varphi_0$:

$$V(\varphi_0+\eta) = V(\varphi_0) + \frac{1}{2}V''(\varphi_0)\eta^2 + \cdots. \tag{1.8}$$

The linear term vanishes at the minimum, and the field equation gives

$$\omega^2 = \boldsymbol{k}^2 + m^2, \qquad m^2 = V''(\varphi_0). \tag{1.9}$$

Mass is therefore the curvature of the potential in the direction of motion. A flat direction has zero curvature and yields a massless mode.

## 1.2 The Broken Complex Scalar

### 1.2.1 Radial and angular motion

Write a complex scalar in Cartesian and polar coordinates:

$$\phi = \phi_R + i\phi_I = \rho e^{i\alpha}, \qquad \phi^* = \phi_R - i\phi_I = \rho e^{-i\alpha}. \tag{1.10}$$

The complex plane is an *internal* field space. At every physical spacetime point $x$, the pair $(\phi_R(x), \phi_I(x))$ specifies one point in that internal plane. Neither internal axis is a direction in the laboratory.

The scalar kinetic term becomes

$$\begin{aligned}\partial_\mu\phi^*\,\partial^\mu\phi &= \partial_\mu\phi_R\,\partial^\mu\phi_R + \partial_\mu\phi_I\,\partial^\mu\phi_I \\ &= \partial_\mu\rho\,\partial^\mu\rho + \rho^2\partial_\mu\alpha\,\partial^\mu\alpha.\end{aligned} \tag{1.11}$$

A globally $U(1)$-invariant potential depends on $\phi^*\phi = \rho^2$, not on $\alpha$. For example,

$$V(\phi) = \lambda\left(\phi^*\phi - \frac{f^2}{2}\right)^2, \qquad \lambda > 0. \tag{1.12}$$

Its minima form a circle. The law is invariant under a common phase rotation, but choosing one equilibrium phase selects one point on the circle.

Set

$$\phi(x) = \frac{f + h(x)}{\sqrt{2}} e^{i\pi(x)/f}. \tag{1.13}$$

Near the selected vacuum, $h$ measures radial displacement and $\pi$ measures tangential displacement with canonical dimensions. Substitution in Eq. (1.11) gives

$$\partial_\mu\phi^*\,\partial^\mu\phi = \frac{1}{2}(\partial_\mu h)^2 + \frac{1}{2}\left(1 + \frac{h}{f}\right)^2(\partial_\mu\pi)^2. \tag{1.14}$$

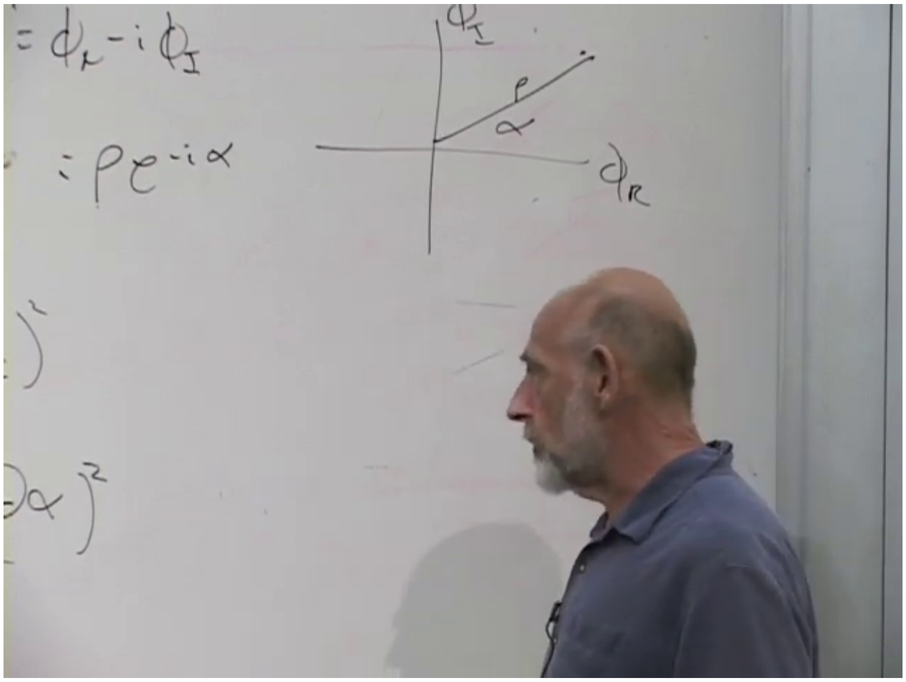


Figure 1.2: The complex scalar represented in its internal $(\phi_R, \phi_I)$-plane. The radius is $\rho$, and $\alpha$ is the internal phase.

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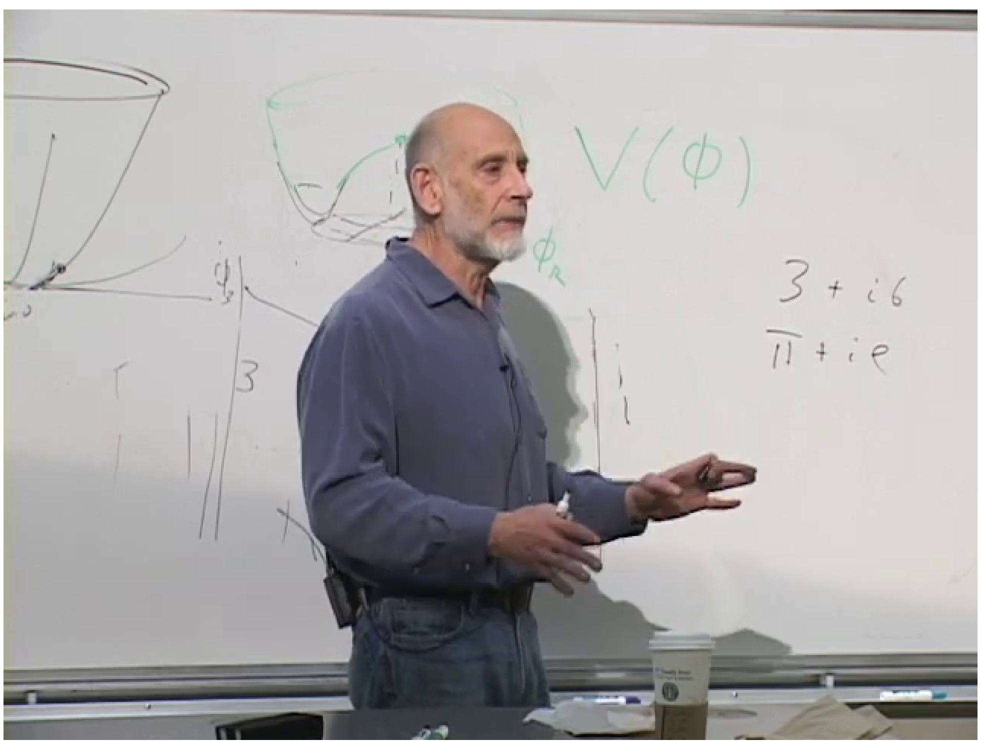


Figure 1.3: The board's radial potential sketches. The potential is symmetric under rotations of the internal field value, even though the field itself need not sit at the origin.

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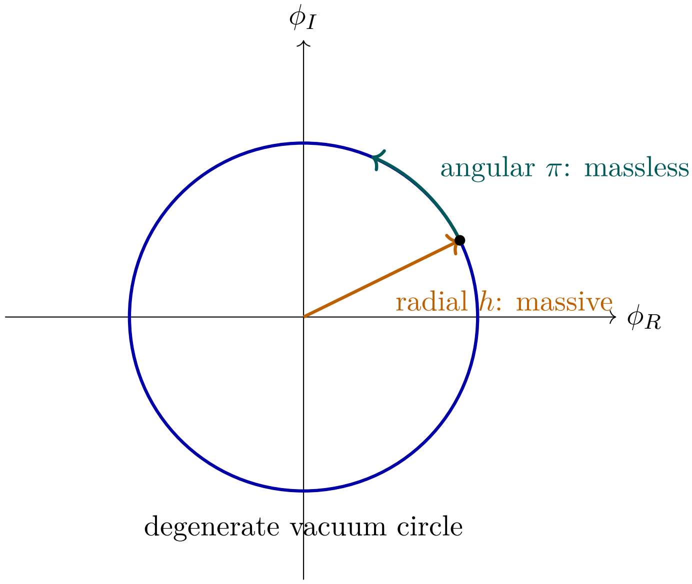


Figure 1.4: The global broken theory has curvature normal to the vacuum circle but no curvature along it. The radial Higgs mode is massive; the angular Goldstone mode is massless.

The potential in Eq. (1.12) supplies

$$m_h^2 = 2\lambda f^2, \qquad m_\pi^2 = 0. \tag{1.15}$$

At energies far below $m_h$, radial motion is expensive, so we may temporarily freeze $h = 0$. The remaining Lagrangian contains only $\frac{1}{2}(\partial\pi)^2$, with no $\pi^2$ term. This is the Goldstone boson.[2]

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**Question from the audience.** Is this the only spontaneous symmetry breaking in the Standard Model?[a]

**Response.** The electroweak Higgs phase is the established symmetry-breaking mechanism relevant here. Its actual symmetry is $SU(2)_L \times U(1)_Y$, not this single $U(1)$, and QCD has additional vacuum structure such as chiral symmetry breaking. The abelian example is a deliberately reduced model of the gauge-boson mechanism.

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[a] Lecture timestamp: 00:16:39.

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**Question from the audience.** Does the Higgs give mass to every particle, including itself?[a]

**Response.** There is a yes-and-no distinction. Electroweak symmetry forbids bare masses for the $W$, $Z$, and Standard Model fermions; their masses appear after the Higgs field takes a nonzero equilibrium magnitude. The radial Higgs mass comes from the curvature of its own potential. Other hypothetical particles can possess symmetry-allowed masses unrelated to this mechanism.

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[a] Lecture timestamp: 00:17:11.

The lecture associates such symmetry-allowed masses with potentially very heavy, difficult-to-observe particles and mentions dark matter as a possible example. That is a motivation, not a mass

[2] Editorial clarification: At two points the spoken explanation reverses the names and says that eating the Goldstone gives the Higgs boson its mass. The board calculation and the remainder of the lecture establish the intended result: radial curvature already gives the Higgs mode a mass, while the gauge boson acquires the Goldstone mode and becomes massive.

prediction. [3]

**Question from the audience.** Is $\rho$ a physical radius, or is it the field value? What exactly is radially symmetric?[a]

**Response.** At each spacetime point the field value is a point in an auxiliary complex plane, and $\rho = |\phi|$ is the distance of that point from the origin. The *potential* is radially symmetric in this internal plane: all points with the same $\rho$ have the same potential energy. This radius is not a distance in ordinary space.

[a]Lecture timestamp: 00:33:08.

**Question from the audience.** Could a sufficiently violent radial kick carry the field over the central maximum to the other side of the potential? [a]

**Response.** In principle a local configuration with enough energy can cross the barrier. Moving a homogeneous macroscopic field over it would require an enormous energy, and a generic violent disturbance would shed energy into many field modes rather than arrive coherently at the opposite point. The low-energy approximation freezes $\rho$ precisely because such radial excursions are inaccessible there.

[a]Lecture timestamp: 00:36:00.

## 1.3 Why the Complex Plane Is Not Spacetime

The distinction between physical and internal coordinates deserves the lecture's full pause. A scalar field is called scalar because it is invariant under rotations of physical coordinate axes. Two real scalar fields $\phi_R(x)$ and $\phi_I(x)$ remain two scalars under those rotations. We may package them into one complex number, but this bookkeeping does not turn them into components of a spatial vector.

A global $U(1)$ transformation rotates the *values* of the field:

$$\phi(x) \longmapsto e^{i\theta}\phi(x), \qquad \rho(x) \longmapsto \rho(x), \qquad \alpha(x) \longmapsto \alpha(x) + \theta. \tag{1.16}$$

The same constant $\theta$ is used at every spacetime point. Consequently, $\partial_\mu \alpha$ is unchanged. This is a symmetry of both the potential and the ordinary kinetic term. Once $\theta$ is allowed to depend on $x$, the derivative detects the variation, and the argument changes.

## 1.4 Making the Phase Rotation Local

### 1.4.1 The ordinary derivative fails

Let

$$\phi'(x) = e^{i\theta(x)}\phi(x). \tag{1.17}$$

Then

$$\partial_\mu \phi' = e^{i\theta}\left(\partial_\mu \phi + i\,\partial_\mu \theta\,\phi\right). \tag{1.18}$$

The extra term proportional to $\partial_\mu \theta$ prevents $\partial_\mu \phi$ from transforming like $\phi$. Therefore $\partial_\mu \phi^* \partial^\mu \phi$ is not invariant under arbitrary local phase rotations.

[3]Editorial clarification: The dark-matter mass scale remains unknown. The lecture's suggestion of a particle around a thousand proton masses is historical speculation, not an experimentally established property of dark matter.

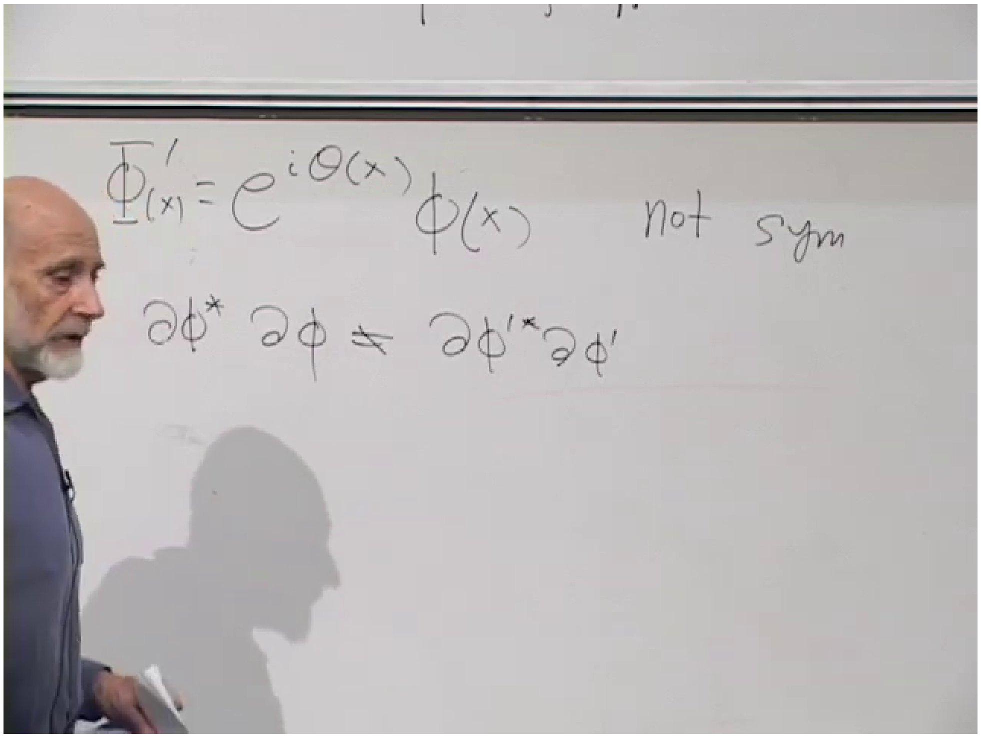


Figure 1.5: The board records the local phase transformation and the failure of the ordinary derivative term to remain invariant.

*Lecture frame: 00:46:49.*

Introduce a vector potential and fix one convention consistently:

$$D_\mu = \partial_\mu + igA_\mu, \qquad A'_\mu = A_\mu - \frac{1}{g}\partial_\mu\theta. \tag{1.19}$$

Now

$$\begin{aligned} D'_\mu\phi' &= \left(\partial_\mu + igA'_\mu\right) e^{i\theta}\phi \\ &= e^{i\theta}\left[\partial_\mu\phi + i(\partial_\mu\theta)\phi + igA_\mu\phi - i(\partial_\mu\theta)\phi\right] \\ &= e^{i\theta} D_\mu\phi. \end{aligned} \tag{1.20}$$

The unwanted terms cancel. The conjugate derivative is

$$(D_\mu\phi)^* = (\partial_\mu - igA_\mu)\phi^*, \tag{1.21}$$

and its opposite phase cancels in the product

$$(D_\mu\phi)^* D^\mu\phi. \tag{1.22}$$

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**Question from the audience.** Was the gauge-field transformation written with the opposite sign in the previous lecture?[a]

**Response.** A simultaneous reversal of the sign in $D_\mu$ and in the transformation law is merely a convention. With $D_\mu = \partial_\mu + igA_\mu$ and $\phi' = e^{i\theta}\phi$, consistency requires $A'_\mu = A_\mu - g^{-1}\partial_\mu\theta$. The alternative convention works equally well if every sign is changed together.

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[a]Lecture timestamp: 00:48:04.

**Question from the audience.** Does the use of an upper-case or lower-case form of $\phi$ on the board mark a different field?[a]

**Response.** No distinction is intended. The changing letter shape is handwriting, not an additional operation or a second scalar. The meaningful distinctions are the prime, the complex-conjugation star, and the spacetime index.

[a]Lecture timestamp: 00:55:02.

**Question from the audience.** Is complex conjugation the same operation as raising or lowering the Lorentz index in the kinetic term? [a]

**Response.** No. The star conjugates the internal complex field. Lorentz indices are raised with the spacetime metric: $D^\mu = g^{\mu\nu} D_\nu$. A Lorentz scalar needs the contraction $(D_\mu\phi)^* D^\mu\phi$; both operations are present and serve different purposes.

[a]Lecture timestamp: 00:56:50.

### 1.4.2 The gauge field has its own dynamics

The field strength

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \tag{1.23}$$

is gauge invariant because the two mixed second derivatives of $\theta$ cancel. Its time–space components contain the electric field, and its space–space components contain the magnetic field. The scalar version of electrodynamics is therefore

$$\mathcal{L} = (D_\mu\phi)^* D^\mu\phi - V(\phi^*\phi) - \frac{1}{4}F_{\mu\nu}F^{\mu\nu}. \tag{1.24}$$

The charged field here is bosonic. It is not the electron field, which is a fermion. A charged helium nucleus or a Cooper-pair condensate illustrates how a bosonic charged degree of freedom can arise.

Before symmetry breaking, the gauge term contains derivatives of $A_\mu$ but no direct mass term. One might try to add

$$\frac{1}{2}m_A^2 A_\mu A^\mu, \tag{1.25}$$

but under Eq. (1.19),

$$A_\mu A^\mu \longmapsto \left(A_\mu - \frac{1}{g}\partial_\mu\theta\right)\left(A^\mu - \frac{1}{g}\partial^\mu\theta\right), \tag{1.26}$$

which is not equal to $A_\mu A^\mu$. A bare gauge-boson mass is forbidden by the local symmetry.

## 1.5 The Higgs Mechanism

### 1.5.1 The covariant derivative produces the mass

Return to the broken potential and use the canonically normalized parametrization

$$\phi(x) = \frac{f + h(x)}{\sqrt{2}} e^{i\pi(x)/f}. \tag{1.27}$$

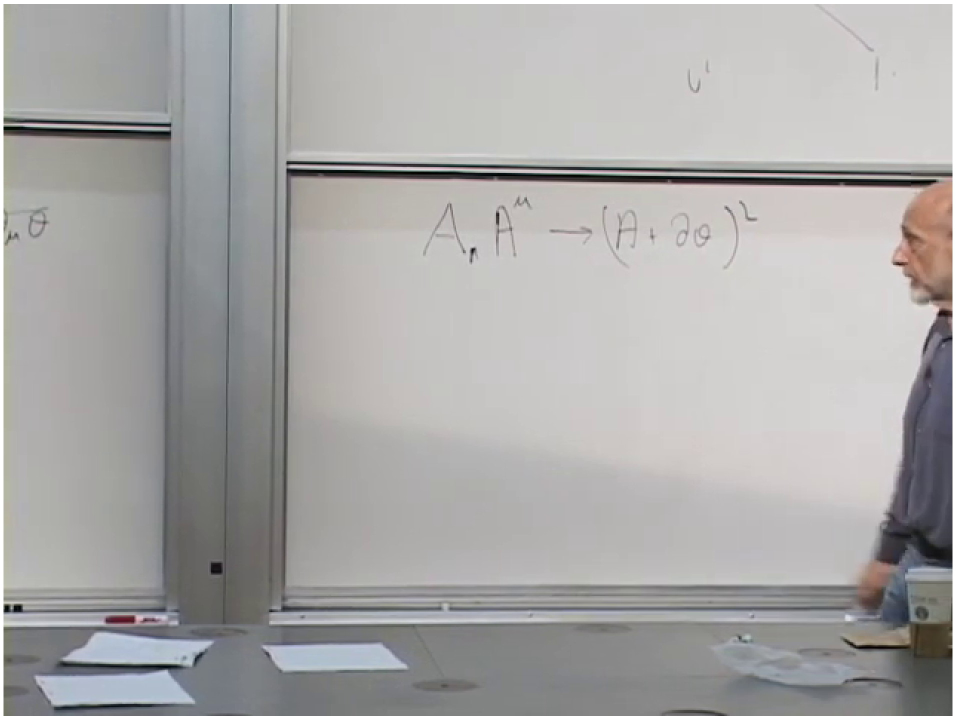


Figure 1.6: A verified board state for the obstruction: under a gauge transformation, $A_\mu A^\mu$ becomes a square containing $\partial_\mu\theta$, so a bare vector mass is not invariant.

*Lecture frame: 01:07:30.*

Its covariant derivative is

$$D_\mu\phi = \frac{e^{i\pi/f}}{\sqrt{2}}\left[\partial_\mu h + i(f+h)\left(\frac{\partial_\mu\pi}{f} + gA_\mu\right)\right]. \tag{1.28}$$

Consequently,

$$(D_\mu\phi)^* D^\mu\phi = \frac{1}{2}(\partial_\mu h)^2 + \frac{1}{2}(f+h)^2\left(\frac{\partial_\mu\pi}{f} + gA_\mu\right)\left(\frac{\partial^\mu\pi}{f} + gA^\mu\right). \tag{1.29}$$

The decisive term appears before any gauge choice:

$$\frac{1}{2}g^2 f^2 A_\mu A^\mu. \tag{1.30}$$

Comparing it with the Proca form $\frac{1}{2}m_A^2 A_\mu A^\mu$, we find

$$\boxed{m_A = gf}. \tag{1.31}$$

The same expansion also produces interactions $g^2 f h A_\mu A^\mu$ and $\frac{1}{2}g^2h^2A_\mu A^\mu$. The vector mass is therefore not an arbitrary violation of gauge symmetry. It comes from a gauge-invariant scalar kinetic term evaluated in a phase with nonzero equilibrium magnitude $f$.

The lecture absorbs $g$ and several factors of $\sqrt{2}$ into its board notation, so it briefly alternates among $f$, $2f$, and $\sqrt{2}f$ while identifying the mass. Those factors have no invariant meaning until the kinetic and potential normalizations are stated. The robust result is $m_A \propto gf$; Eq. (1.31) gives the factor for the convention used here.

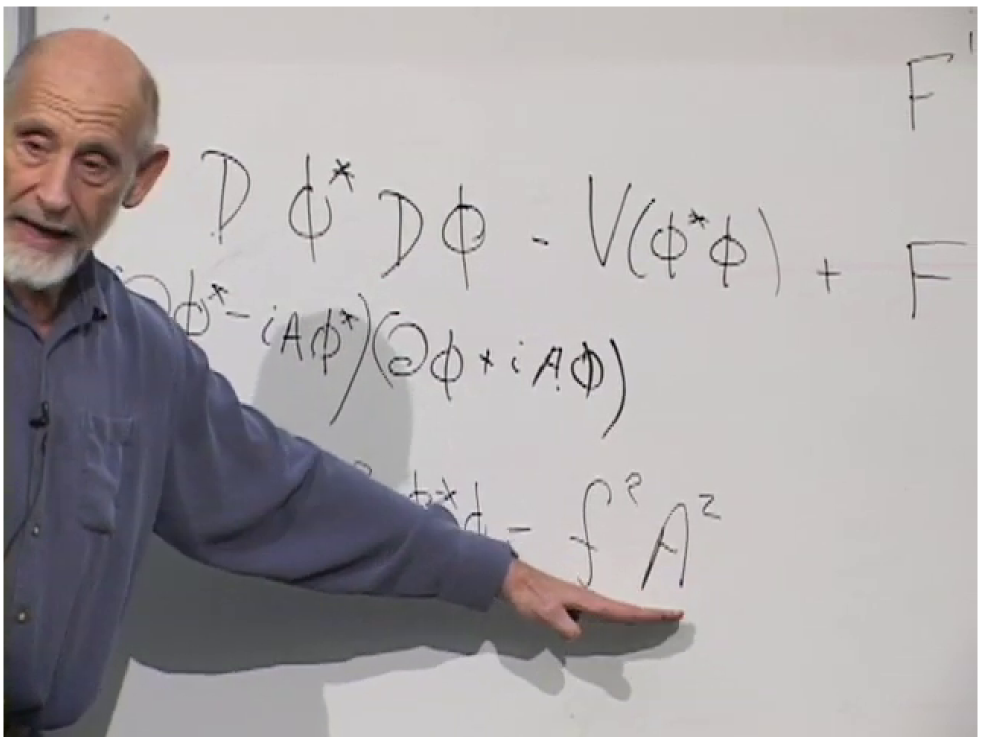


Figure 1.7: The board isolates the $f^2A^2$ contribution inside $(D\phi)^*(D\phi)$. Numerical factors depend on how the scalar field and the gauge coupling are normalized; with Eqs. (1.13) and (1.19), the result is $m_A = gf$.

*Lecture frame: 01:10:10.*

### 1.5.2 The phase is absorbed

Under a gauge transformation the phase variable changes as

$$\pi'(x) = \pi(x) + f\,\theta(x). \tag{1.32}$$

Choose

$$\theta(x) = -\frac{\pi(x)}{f}. \tag{1.33}$$

Then $\pi' = 0$. This is unitary gauge, in which

$$\phi'(x) = \frac{f + h(x)}{\sqrt{2}} \tag{1.34}$$

is real, and Eq. (1.29) reduces to

$$(D_\mu \phi')^* D^\mu \phi' = \frac{1}{2}(\partial_\mu h)^2 + \frac{1}{2} g^2 (f+h)^2 A'_\mu A'^\mu. \tag{1.35}$$

The angular Goldstone field has disappeared from the list of independent variables, while the vector mass remains.

The lecture first demonstrates this in the low-energy limit $\rho = f$, where

$$\begin{aligned} \phi &= f e^{i\alpha}, \\ D_\mu \phi &= i f e^{i\alpha} \left(\partial_\mu \alpha + g A_\mu\right), \\ (D_\mu \phi)^* &= -i f e^{-i\alpha} \left(\partial_\mu \alpha + g A_\mu\right). \end{aligned} \tag{1.36}$$

Their product is

$$f^2 \left(\partial_\mu \alpha + g A_\mu\right) \left(\partial^\mu \alpha + g A^\mu\right). \tag{1.37}$$

Choosing $\theta = -\alpha$ leaves $g^2 f^2 A'_\mu A'^\mu$.

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**Question from the audience.** Can the whole Lagrangian be rewritten in variables that make the disappearance of the angular mode explicit? [a]

**Response.** Yes. In polar variables the phase appears only through the gauge-invariant combination $\partial_\mu\alpha + gA_\mu$. A gauge transformation with $\theta = -\alpha$ sets the phase to zero everywhere that this gauge is well-defined. The scalar kinetic term then becomes a vector mass term, while the gauge-invariant field-strength term is unchanged.

[a] Lecture timestamp: 01:13:49.

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**Question from the audience.** Are oscillations in $\alpha$ indistinguishable from gauge oscillations? [a]

**Response.** In the gauged theory, a pure change of $\alpha$ that is compensated by the corresponding change of $A_\mu$ is a gauge redescription, not a distinct physical wave. The invariant quantity is $\partial_\mu\alpha + gA_\mu$. The physical mode that was represented by the global Goldstone field reappears as the longitudinal polarization of the massive vector.

[a] Lecture timestamp: 01:18:02.

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It is common to say that a local gauge symmetry is spontaneously broken. Operationally this language is useful: after choosing a gauge, the scalar field has a nonzero vacuum value and the

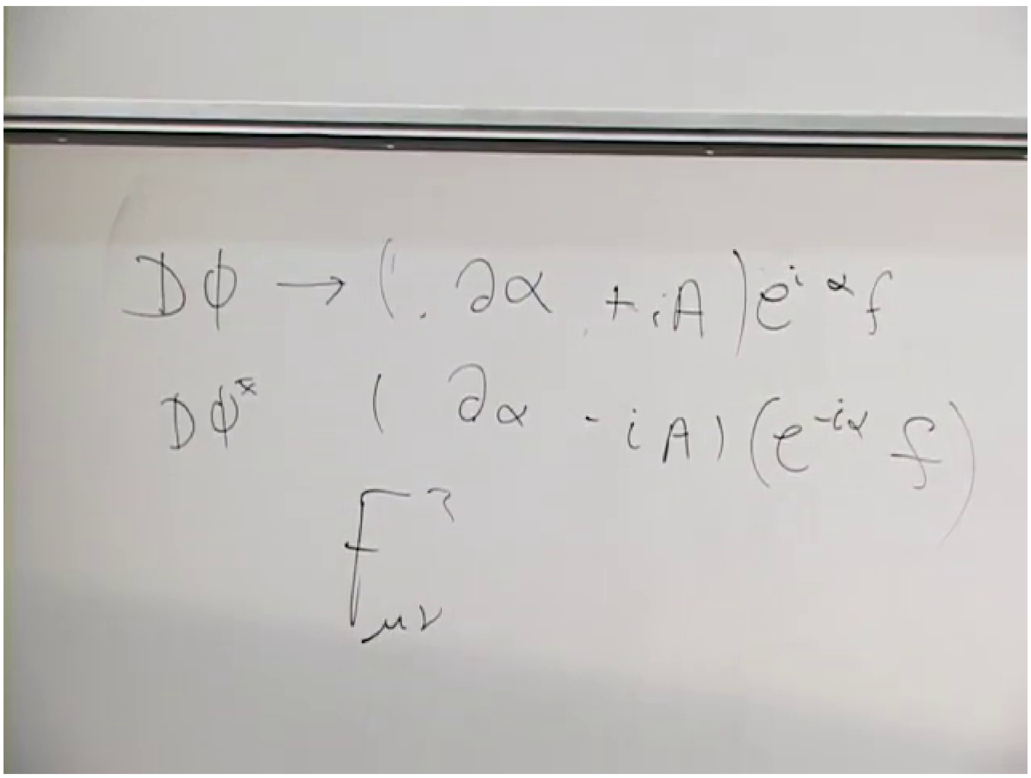

Figure 1.8: The complete frozen-radius board state: the covariant derivatives contain the phase derivative and vector potential together. The lecture next chooses $\theta = -\alpha$, eliminating the phase and leaving the vector mass term.

*Lecture frame: 01:16:20.*

spectrum contains a massive vector. Strictly, however, gauge transformations relate redundant descriptions, not distinct physical states. [4]

## 1.6 Where the Goldstone Degree of Freedom Went

Nothing physical can vanish merely because we changed variables. Count the degrees of freedom before and after entering the Higgs phase.

Beforehand:

$$\underbrace{2}_{\text{massless vector}} + \underbrace{2}_{\text{complex scalar}} = 4. \tag{1.38}$$

A massless vector has two transverse polarizations. The complex scalar has two real components, radial and angular.

Afterward:

$$\underbrace{3}_{\text{massive vector}} + \underbrace{1}_{\text{radial Higgs}} = 4. \tag{1.39}$$

A massive vector can be brought to rest. In its rest frame no spatial direction is privileged, so it must have three polarization states. When it moves, two are transverse and one is longitudinal. The Goldstone mode has become that longitudinal polarization.

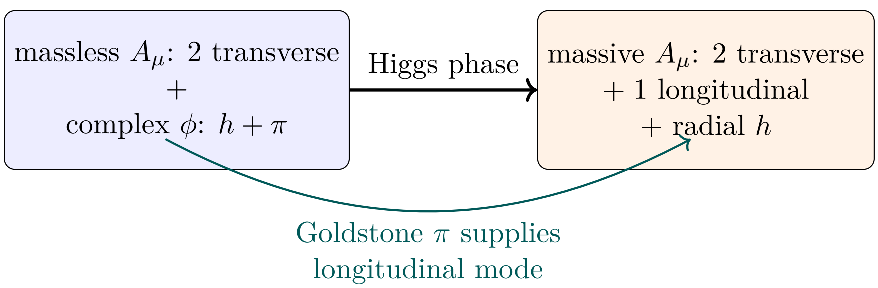


Figure 1.9: The Higgs mechanism rearranges, rather than destroys, physical degrees of freedom.

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**Question from the audience.** How can the angular degree of freedom disappear without violating degree-of-freedom counting?[a]

**Response.** It does not disappear physically. A massless vector has two helicities because it cannot be brought to rest. A massive vector has three polarizations, including a longitudinal one. The Goldstone mode becomes that third polarization, leaving the total count unchanged.

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[a] Lecture timestamp: 01:19:13.

The lecture offers a useful kinematic test. Imagine a vector particle with mass, slow it to rest, and accelerate it in a new direction. A polarization that was transverse to the original motion can become longitudinal relative to the new motion. Likewise, if a circularly polarized massive particle is carried through rest and sent in the opposite direction, its helicity label reverses. A strictly massless particle cannot be taken through this experiment, because no inertial frame can bring it to rest. Its two helicities therefore form a consistent physical spectrum.

[4] Editorial clarification: A local gauge redundancy is not broken in the same gauge-invariant sense as a global symmetry. The precise statement is that the theory is in a Higgs phase. Gauge-fixed perturbation theory describes that phase by a nonzero scalar expectation value and the resulting massive vector spectrum.

## 1.7 The Superconducting Analogue

In a conventional superconductor, electrons form Cooper pairs. A pair is a boson carrying electric charge $2e$, and the condensate is described by a charged complex order parameter. When its magnitude is nonzero, the combination of its phase and the electromagnetic vector potential produces the Anderson–Higgs mechanism. Static magnetic fields decay over the London penetration depth rather than extending freely through the bulk. In this precise medium-dependent sense, the photon behaves as though it has acquired a mass.

This example also clarifies the opening distinction. The electromagnetic field is not permanently converted into a different elementary particle. The massive propagation law belongs to the collective state inside the material; outside, the vacuum photon remains massless.

## 1.8 What Has Been Established

The argument can now be compressed without losing its logic:

1. A broken global continuous symmetry has a flat angular direction and therefore a massless Goldstone mode.
2. Local phase invariance requires a gauge potential and replaces $\partial_\mu$ by $D_\mu$.
3. A bare $A_\mu A^\mu$ mass violates the local transformation law.
4. When the charged scalar has nonzero equilibrium magnitude $f$, the gauge-invariant term $|D_\mu\phi|^2$ contains $\frac{1}{2}g^2 f^2 A_\mu A^\mu$.
5. The phase can be removed by a gauge choice, and its physical degree of freedom becomes the longitudinal polarization of the massive vector.
6. The radial Higgs excitation remains, with mass fixed by the curvature of the scalar potential.

The next task is to give fermions their masses. Electrons and quarks cannot use the vector mechanism directly; their Higgs couplings are Yukawa interactions. Once the scalar settles at $f$, those interactions become fermion mass terms.

The broad statement that familiar particles would be massless without electroweak symmetry breaking requires one final qualification. The elementary quark and charged-lepton masses, and the $W$ and $Z$ masses, do vanish with the Higgs expectation value. Most of the proton and neutron mass, however, arises dynamically from QCD confinement and strong-field energy rather than from the quarks' Higgs masses. That distinction does not alter the Higgs mechanism derived here; it keeps its domain of application precise.